Representation and Modeling of 3D Objects (P1)

Contents

- **Implicit functions**
- **Polygonal representation**
- **Bicubic parametric patches**
- **Constructive solid geometry**
- **Space subdivision techniques**
- Procedural modeling
- Artificial intelligence

Motivation

Object representation and modeling determine:

- Data structure
- Processing algorithm
- Design of software packages and hardware modules
- Resources required through 3D processing pipeline
- Final appearance of the object
- Approach for editing the shape of the object

Object representation

1. Implicit functions
   - object is an analytical definition, e.g., a sphere is defined as $x^2+y^2+z^2=r^2$
2. Polygonal
   - objects are approximated by a set of planar polygonal facets
3. Bicubic parametric patches
   - objects are represented as a set of patches. The sets of patches are polynomials in two parametric variables (e.g., Bézier, Hermite, B-spline, etc.)
4. Constructive solid geometry (CSG)
   - object is a collection of elementary objects such as spheres, cylinders and boxes
5. Space subdivision techniques
   - object embedded in a space where the zones are labeled according to object occupancy
6. Procedural modeling

Implicit functions

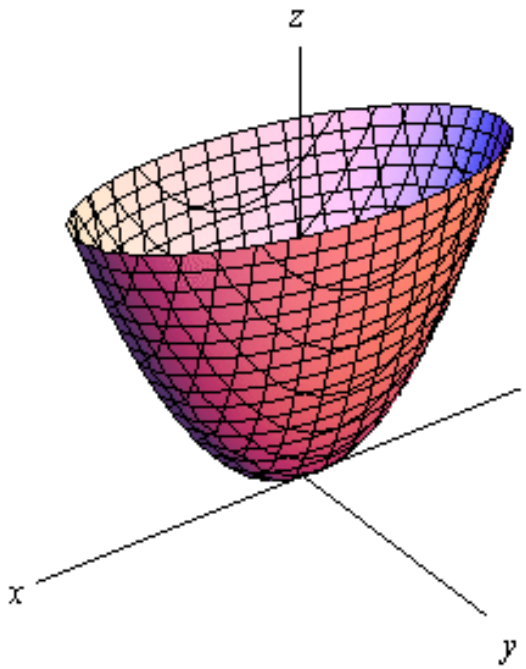
- Some surfaces can be represented as points of functions (defined over 3D space)
  - Places where a function f(x,y,z)=0
- Some objects are easy represent this way
  - Spheres, ellipses, and similar
  - More generally, quadratic surfaces

    $F(x,y,z)=ax^2 + by^2 + cz^2 + 2dxy + 2eyz+ 2fxz + 2gx + 2hy + 2jz + k = 0$
  - Shapes depends on all the parameters a,b,c,…,k

Quadratic surfaces

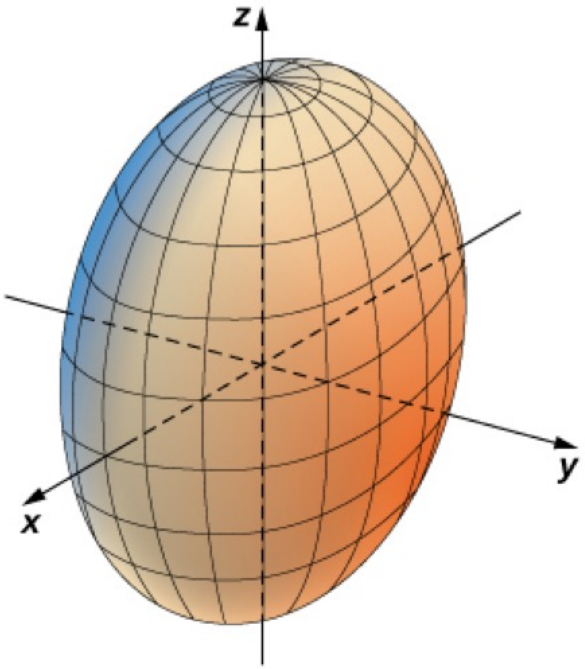
Elliptical paraboloid

$$\frac{z}{c} = \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2$$



Ellipsoid

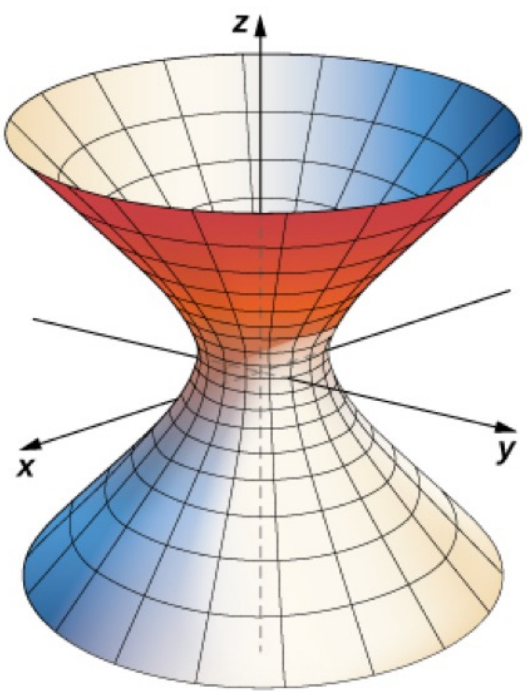
$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$



Quadratic surfaces

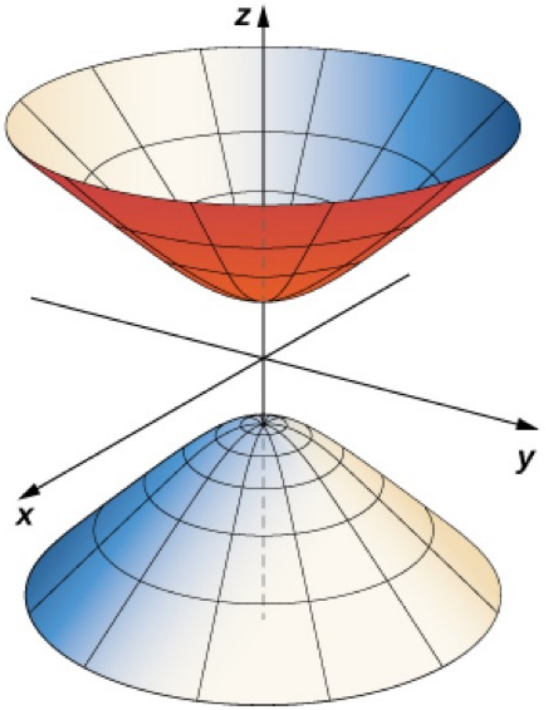
Hyperboloid of One Sheet

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$



Hyperboloid of Two Sheets

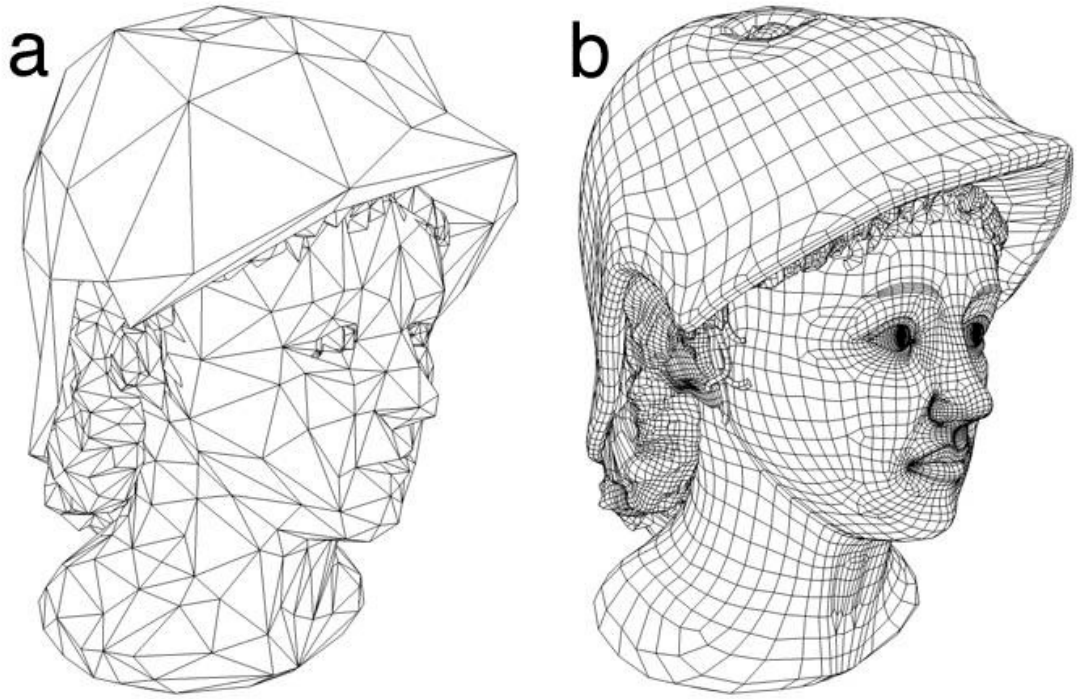
$$\frac{z^2}{c^2} - \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$



Polygonal representation

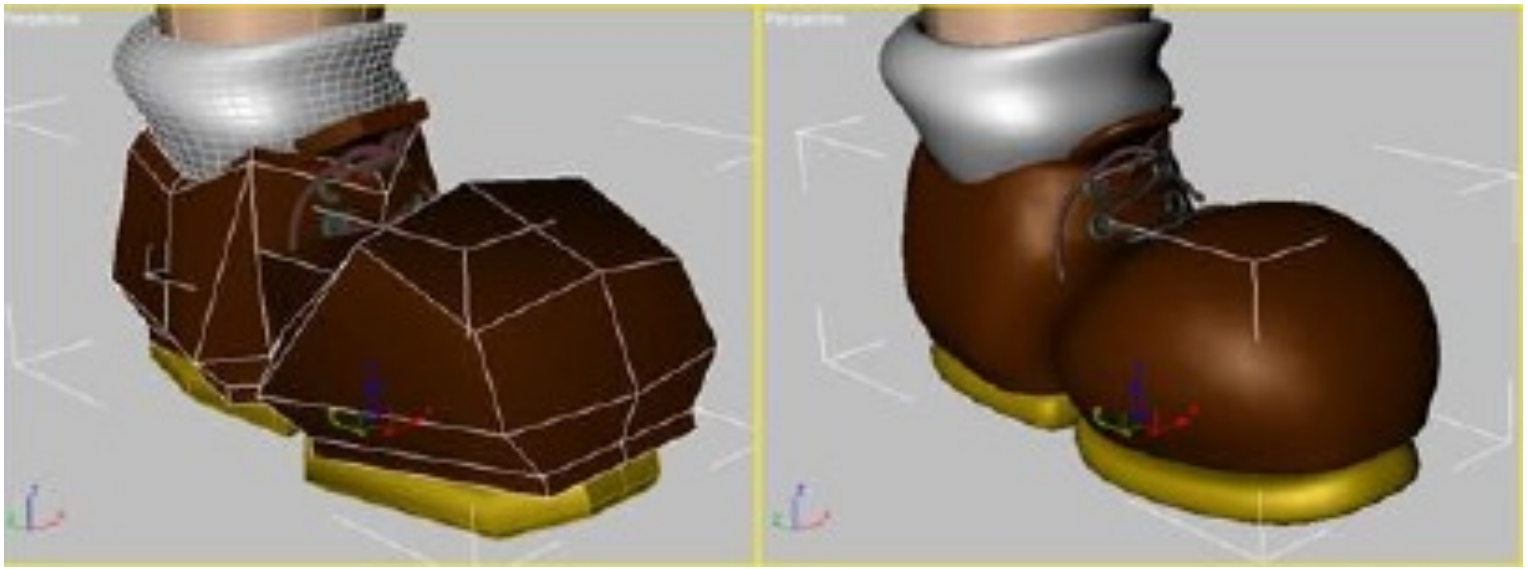
- **Polygonal mesh**, called **Boundary representation** (**Brep**)

  Geometric and topological description of the boundary or surface of the object

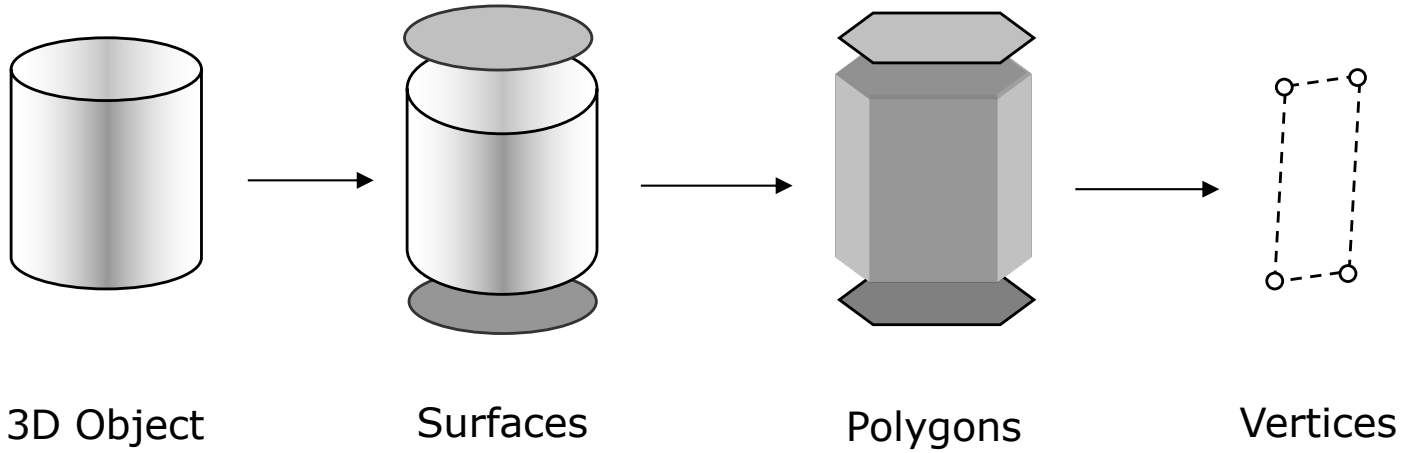


Polygon based modeling

Images taken from the 3D Studio Max tutorial of Modeling Clothing



Polygonal representation



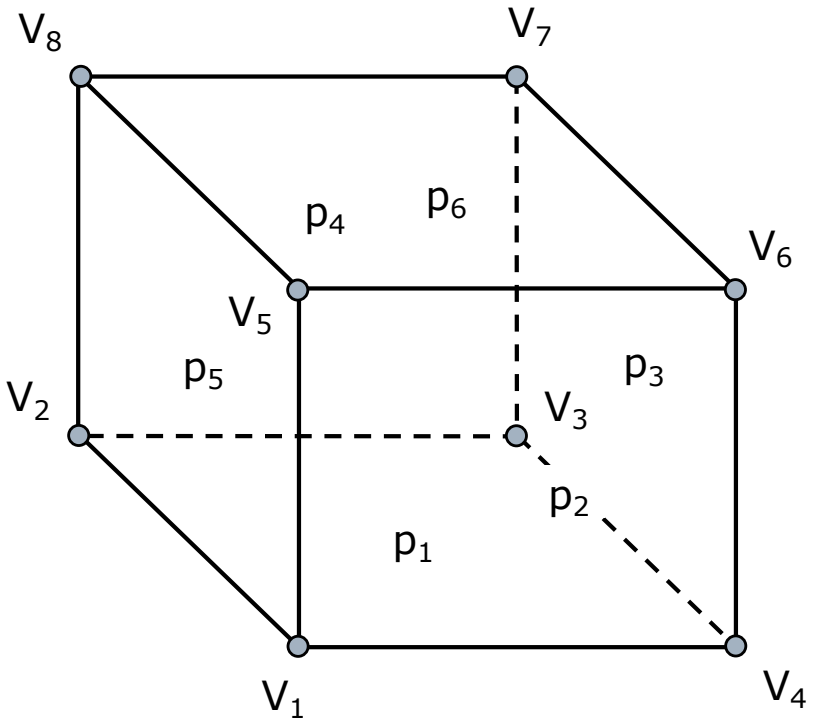
Hierarchical polygonal decomposition of a 3D object

Polygonal representation

- Advantages
  - Ubiquitous in computer graphics
  - Simple and straightforward
  - Used as intermediate form for many other data structures
- Issues
  - Approximation of curved surfaces
  - Accuracy depends extremely on the number of polygons

    e.g., zoom, scale
  - Texture mapping
  - Smooth surfaces
  - Shading algorithms

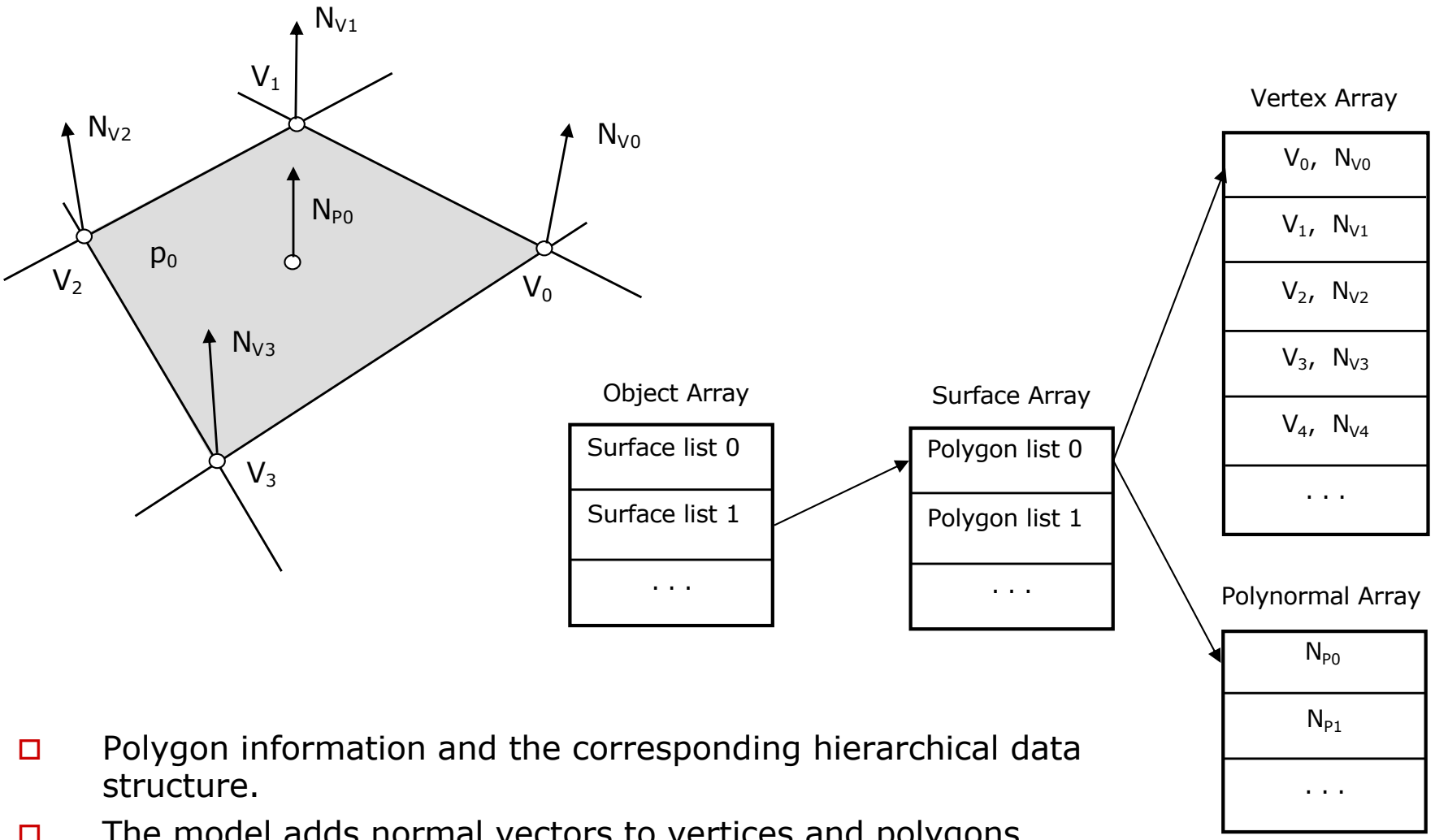
Brep – vertex based model



Polygon (face)	Vertices
p_1	$V_1V_2V_3V_4$
p_2	$V_6V_5V_1V_4$
p_3	$V_7V_6V_4V_3$
p_4	$V_8V_7V_3V_2$
p_5	$V_5V_8V_2V_1$
p_6	$V_8V_5V_6V_7$

Vertex	Coordi-nates
V_1	$x_1\ y_1\ z_1$
V_2	$x_2\ y_2\ z_2$
V_3	$x_3\ y_3\ z_3$
V_4	$x_4\ y_4\ z_4$
V_5	$x_5\ y_5\ z_5$
V_6	$x_6\ y_6\ z_6$
V_7	$x_7\ y_7\ z_7$
V_8	$x_8\ y_8\ z_8$

Brep - polygon based model

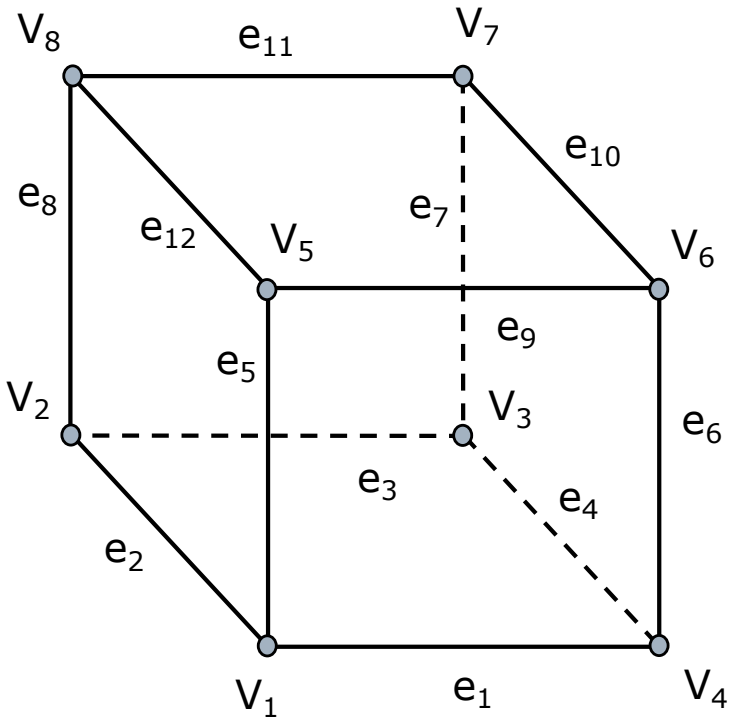


- Polygon information and the corresponding hierarchical data structure.
- The model adds normal vectors to vertices and polygons.

Polygonal model - Data structure

- The data structure consists of four arrays:
  1. *Object Array* - an array of all surfaces of the 3D object modelled through polygons. A position of the array is a list of indices of the polygonal components.
  2. *Surface Array* - it is the set of polygons that describes the surface. A position in the array describes a polygon as a list of indices of the polygon vertices and an index of the polygon normal.
  3. *Vertex Array* - describes the information that is associated to each vertex. Three coordinates and a normal vector describe a vertex. In fact the normal vector has three components: its projections onto the coordinate axes - Nx, Ny and Nz.
  4. *Polynormal Array* - a set of normal vectors of the polygons. Each vector has three vector components.
- Disadvantage:
  - Edges are shared between adjacent polygons. The edge is not explicitly contained in the data structure.
  - An edge is processed twice.

Brep – edge based model

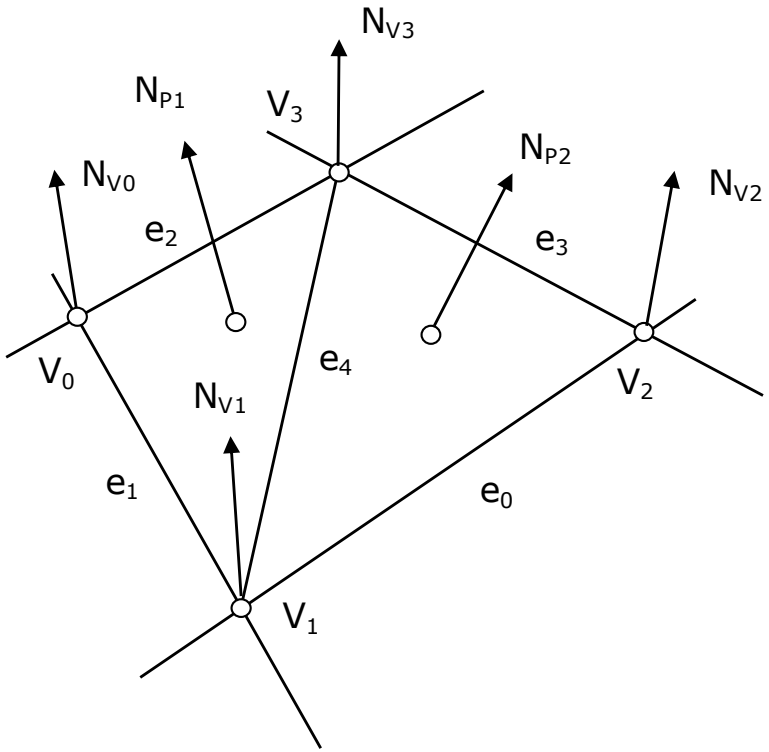


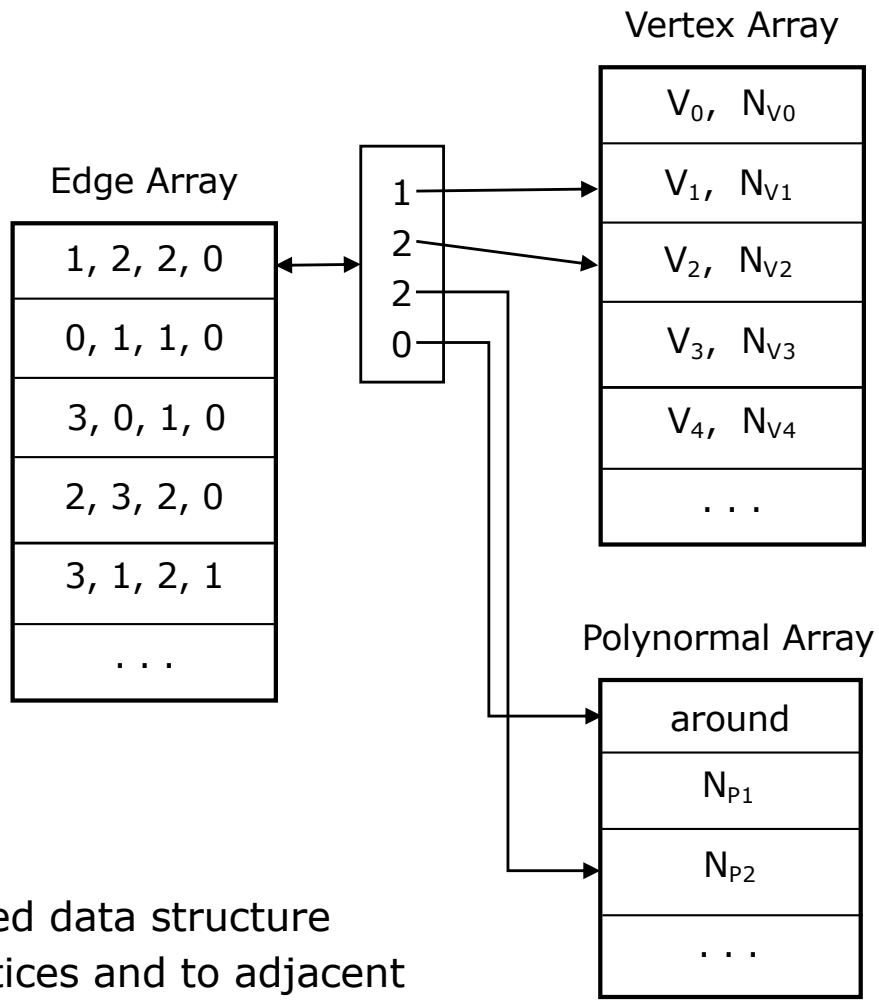
Polygon (face)	Edges
p_1	$e_1e_2e_3e_4$
p_2	$e_1e_6e_9e_5$
p_3	$e_4e_7e_{10}e_6$
p_4	$e_5e_8e_{11}e_7$
p_5	$e_2e_5e_{12}e_8$
p_6	$e_9e_{10}e_{11}e_{12}$

Edge	Vertices
e_1	$V_4\ V_1$
e_2	$V_1\ V_2$
e_3	$V_2\ V_3$
e_4	$V_3\ V_4$
e_5	$V_5\ V_1$
e_6	$V_4\ V_6$
e_7	$V_3\ V_7$
e_8	$V_2\ V_8$
e_9	$V_5\ V_6$
e_{10}	$V_6\ V_7$
e_{11}	$V_7\ V_8$
e_{12}	$V_8\ V_5$

Vertex	Coordi-nates
V_1	$x_1\ y_1\ z_1$
V_2	$x_2\ y_2\ z_2$
V_3	$x_3\ y_3\ z_3$
V_4	$x_4\ y_4\ z_4$
V_5	$x_5\ y_5\ z_5$
V_6	$x_6\ y_6\ z_6$
V_7	$x_7\ y_7\ z_7$
V_8	$x_8\ y_8\ z_8$

Improved edge based model





- Polygon information and the edge based data structure
- The model adds normal vectors to vertices and to adjacent polygons

Creating the Surface Points

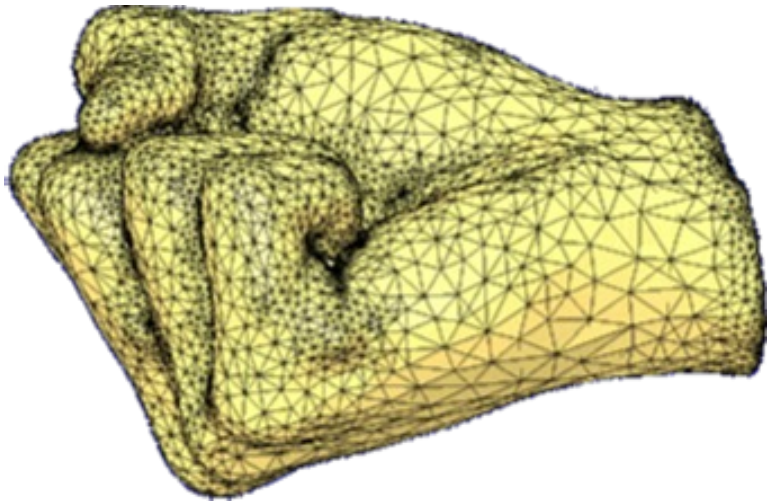
- Approaches:
  - Manual placement
  - Mathematical (geometrical) generation
    - Implicit functions
    - Swapping, extrusion
  - Scanning real objects
    - Laser scanning
    - Tomographic methods
    - Computer vision

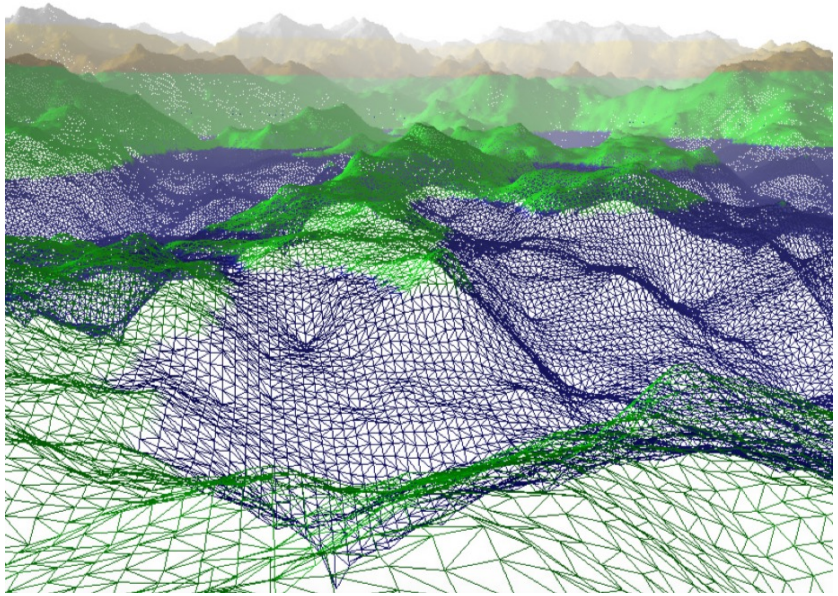




Triangulation

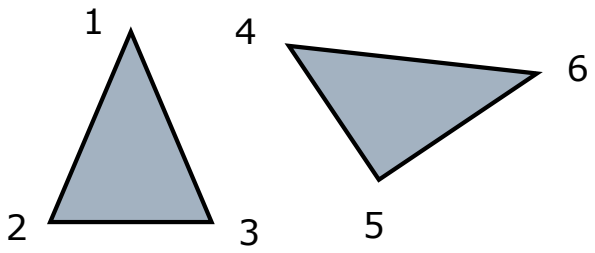
- Triangles are the first choice of polygons
  - Triangles are always planar
  - Triangles are always convex polygons -> simple and fast algorithms
  - Graphics hardware is often optimized for triangles
    - Ex. Nvidia processes nx10M triangles per second, and nx100M vertices per second
    - 1080 GTX, processes $16x10^3$M triangles per second





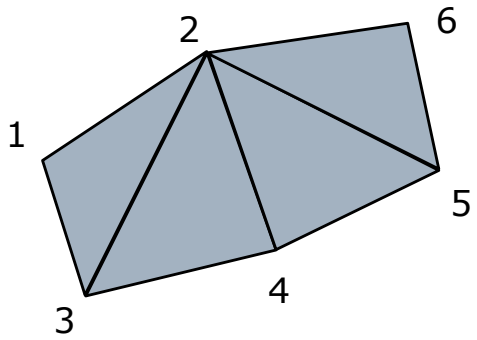
Triangle representation

- Lists



3 vertices per triangle

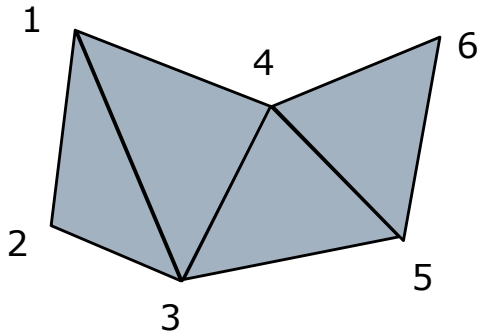
- Fans



6/4=1.5 vertices per triangle

- Strips

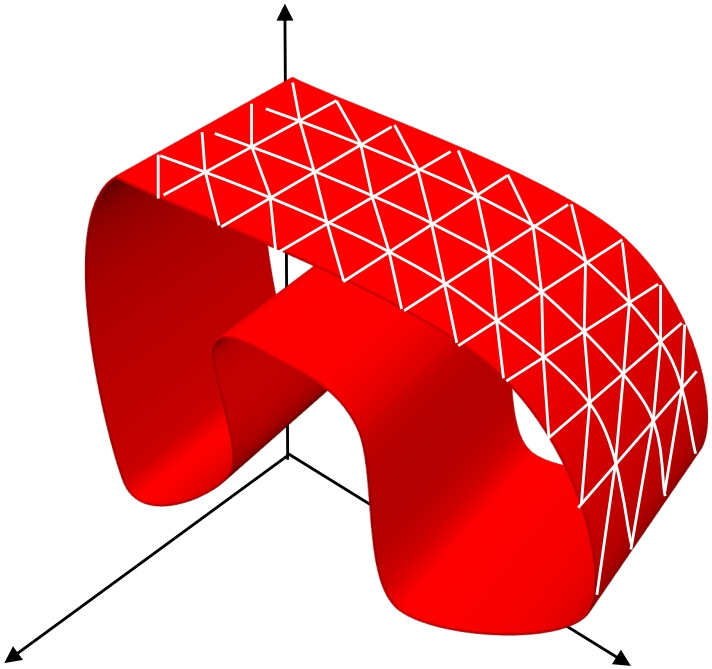
  the most efficient representation

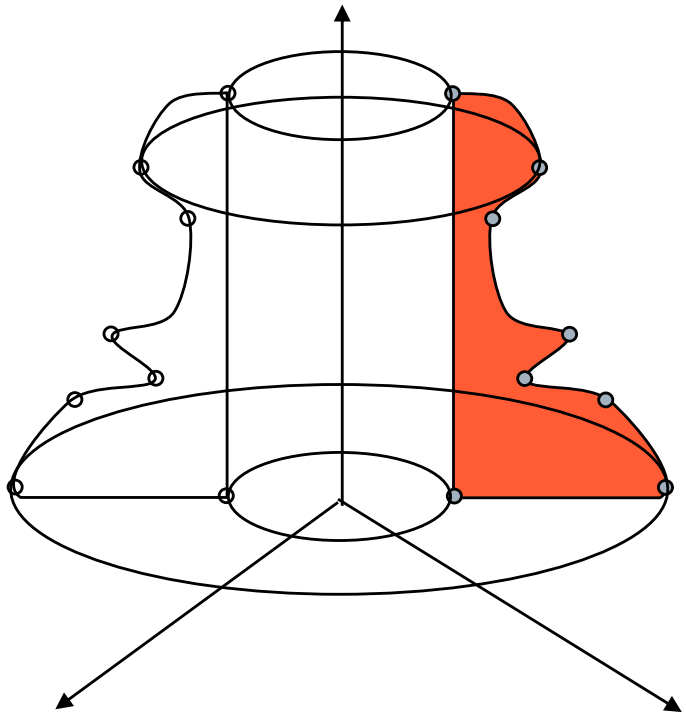


6/4=1.5 vertices per triangle

Polygonal objects by sweeping

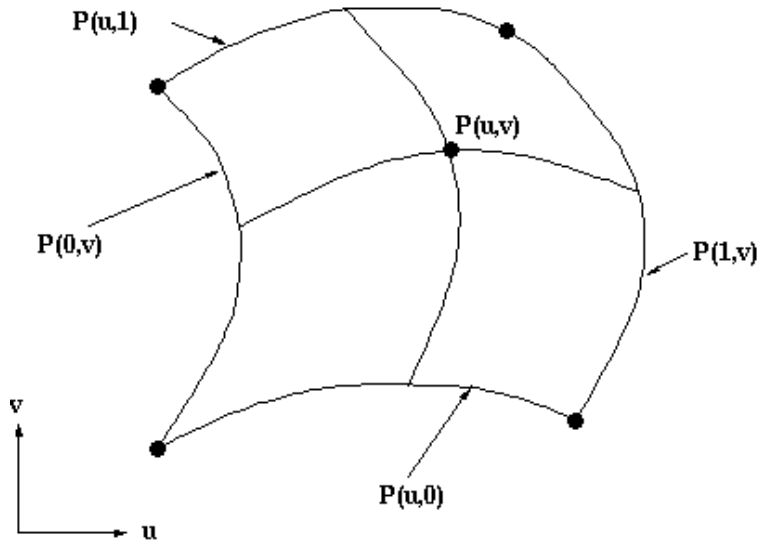
- Sweep a cross section along a curve
- The cross section can be allowed to vary in shape as it is swept
- Define intervals along the curve at which the cross section is to be placed
- Join vertex to vertex with the previous emplacement of the cross section, with straight lines



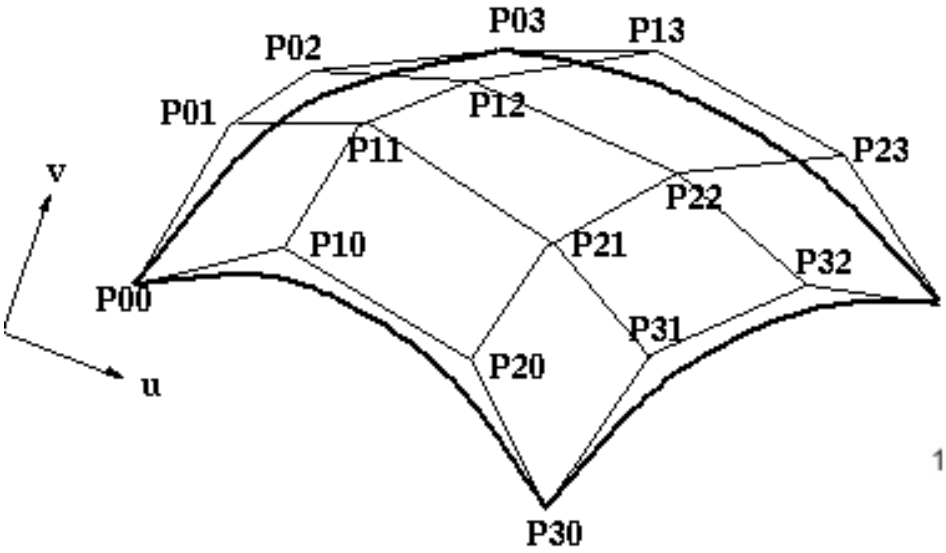


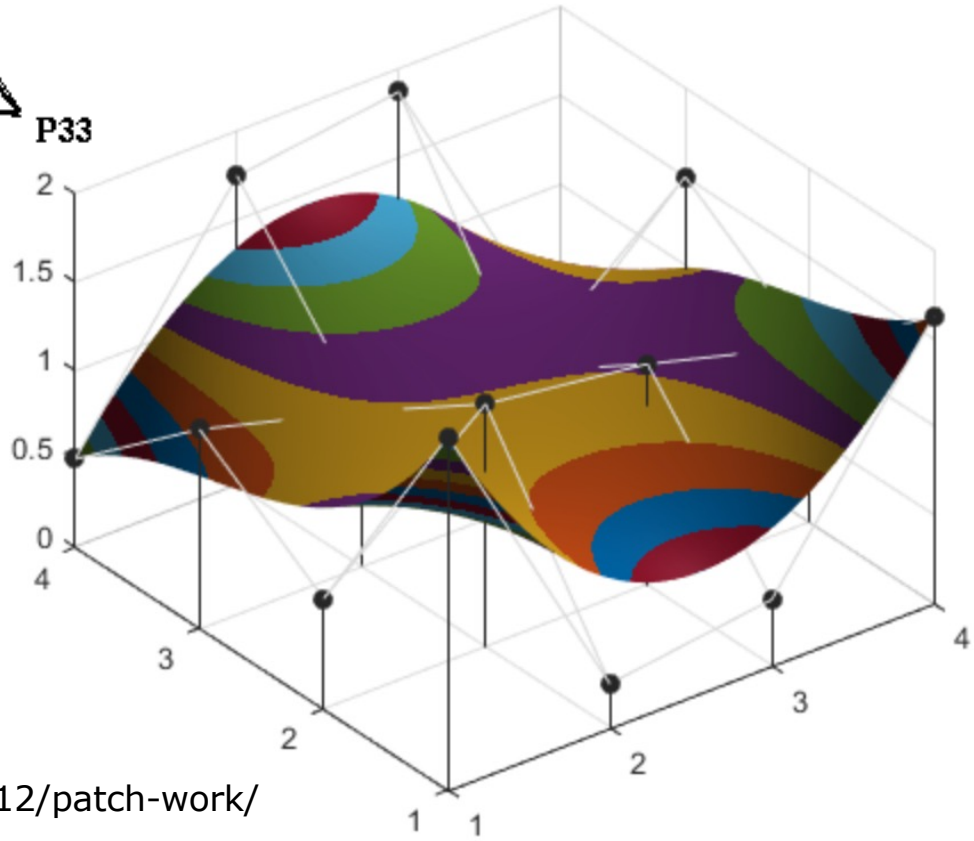
Bicubic parametric patches

- $P(u,v)=[x(u,v), y(u,v), z(u,v)]$, $0 \le u \le 1$, $0 \le v \le 1$
- Curves:
  - Hermite
  - Bézier
  - Spline
  - Uniform B-Spline
- Surfaces
  - Hermite
  - Bézier
  - B-Spline
- Advantages:
  - Parametric representation is exact and economic since it is analytical. With a polygonal object, the shape accuracy can only be approximated at the expense of extra processing and database costs
  - Deformations of parametric surfaces appear smooth. This is not generally the case with a polygonal object



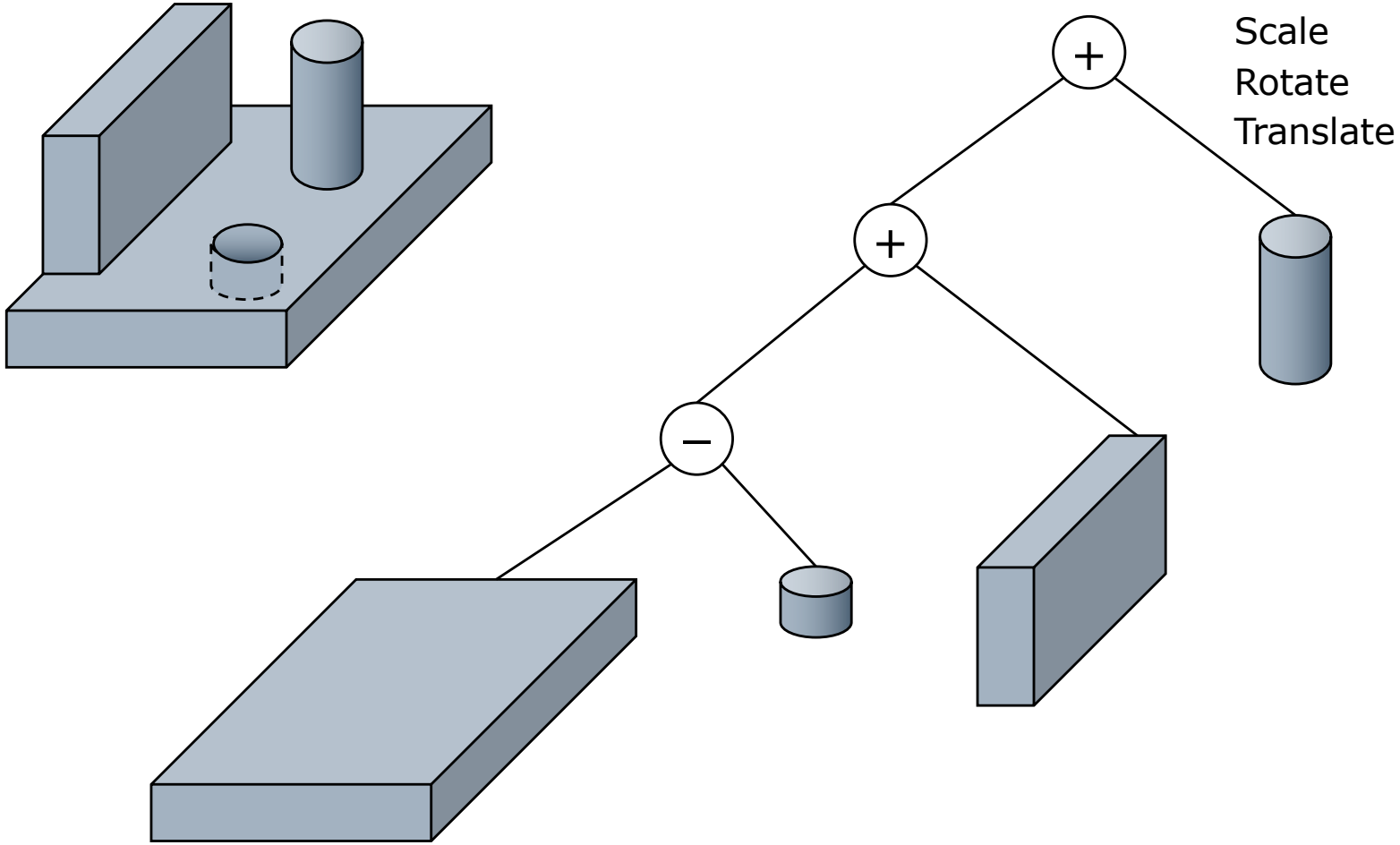
Bicubic parametric patches



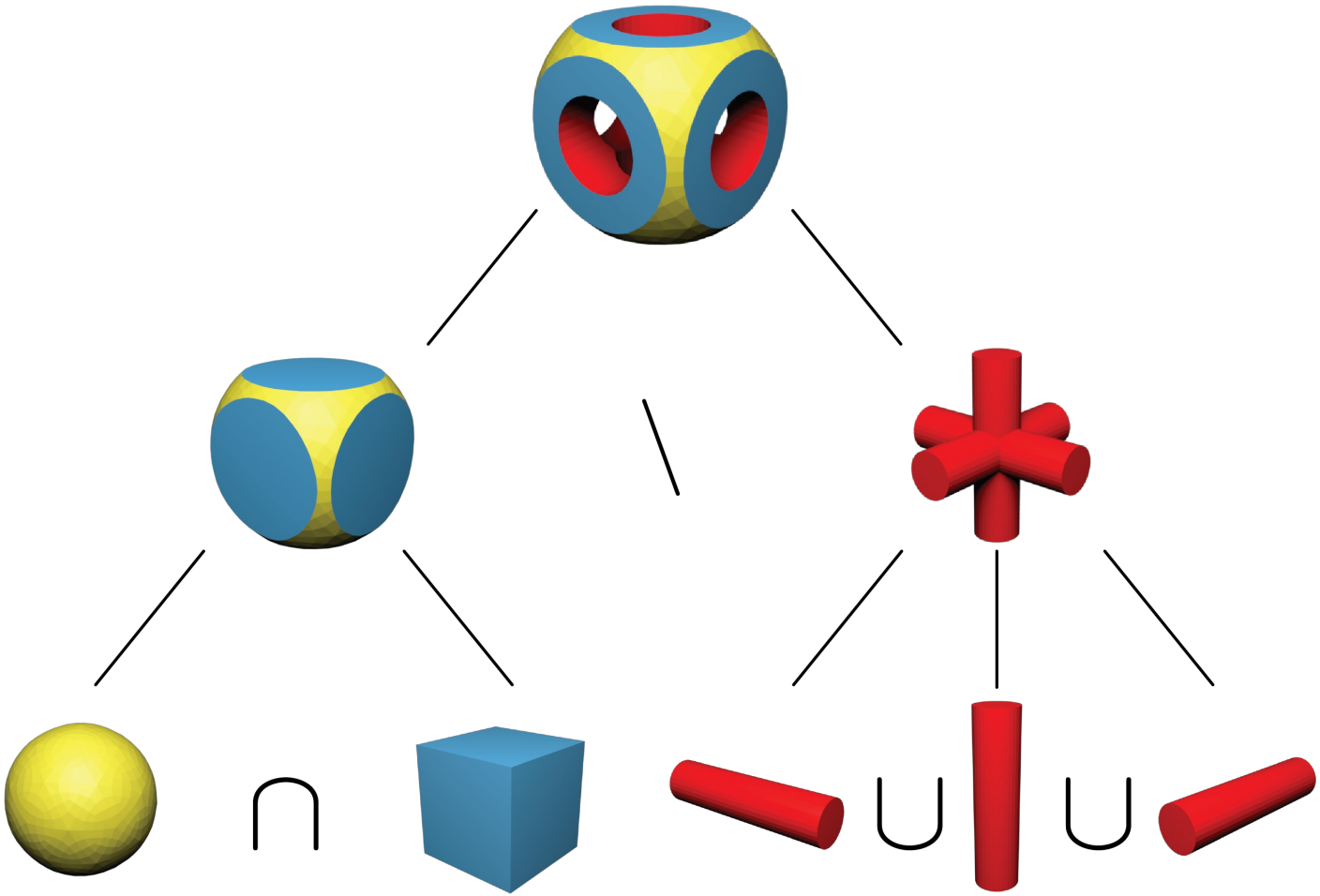


https://blogs.mathworks.com/graphics/2015/05/12/patch-work/

Constructive Solid Geometry (CSG)

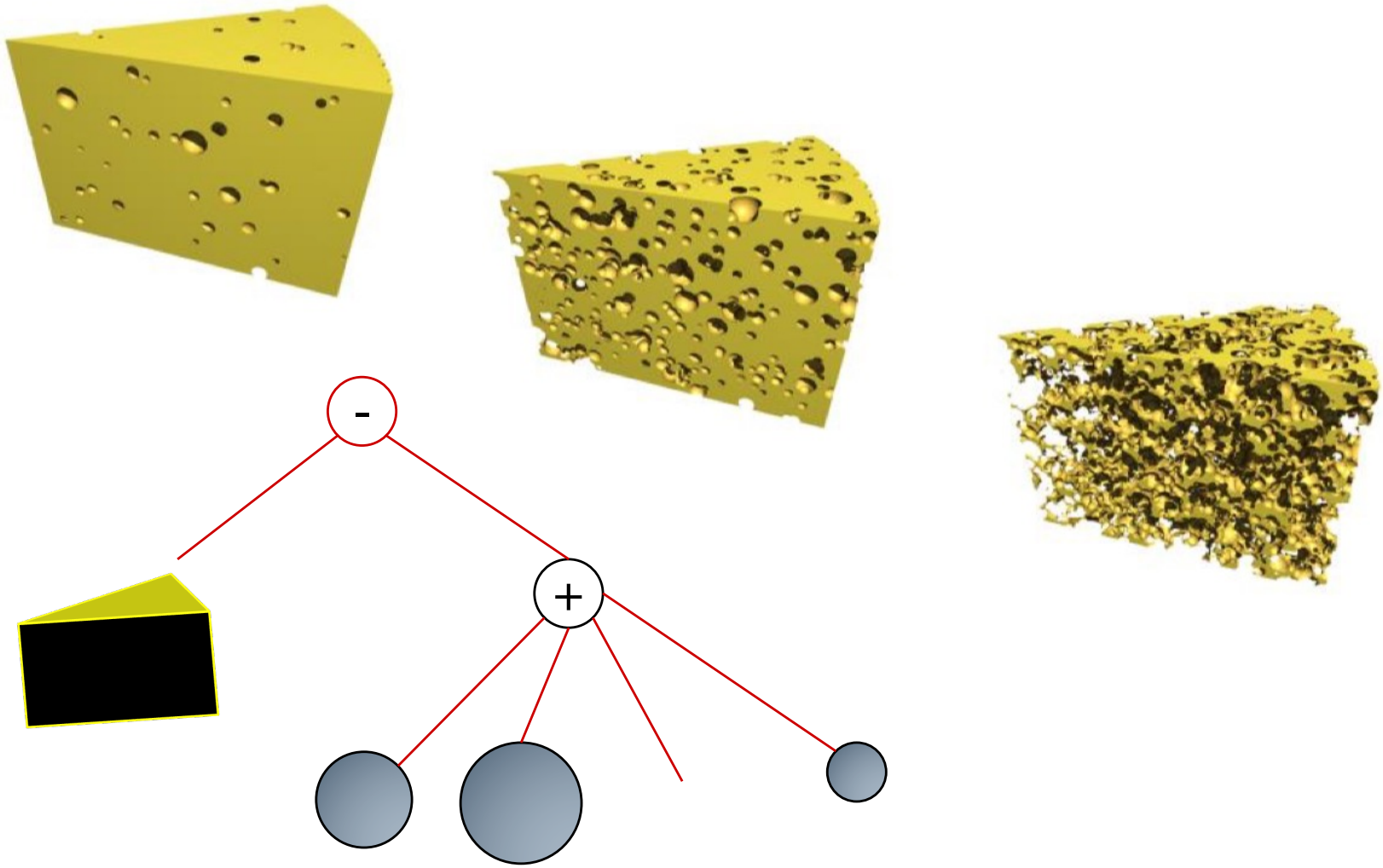


Constructive Solid Geometry (CSG)



https://pymesh.readthedocs.io/en/latest/mesh_boolean.html

CSG Objects



Constructive Solid Geometry (CSG)

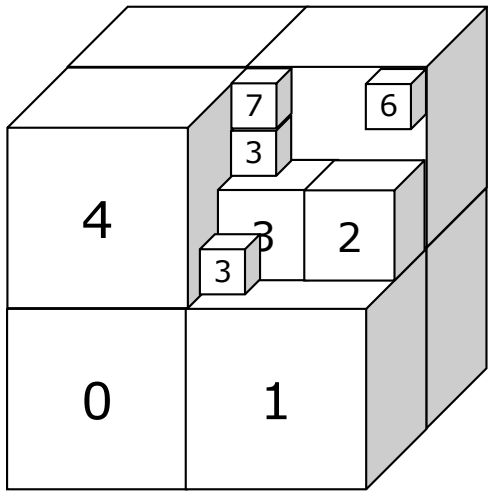
- Hierarchical modeling based on tree structure
  - The internal nodes are set to operations: union, intersection or difference (sometimes complement)
  - The edges of the tree have transformations associated with them

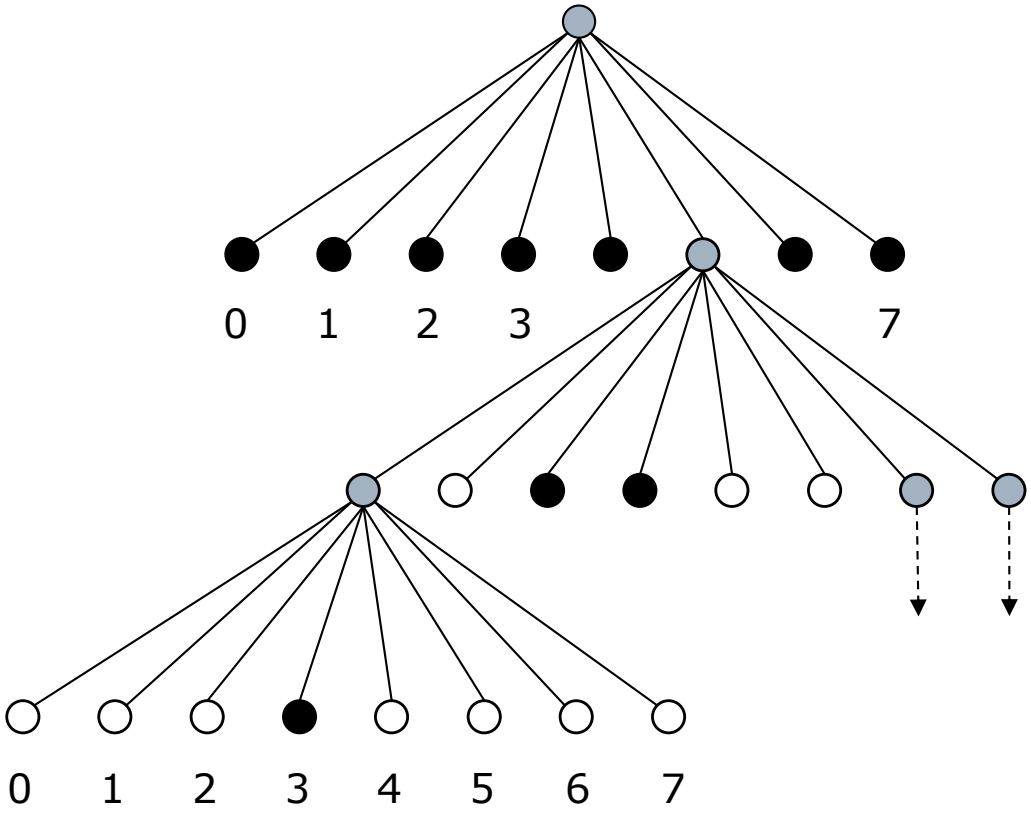
    Scale, rotate, translate
  - The leaves contain only geometry
- Allows complex shapes with only a few primitives
  - cylinders, cubes, etc, or quadric surfaces
- Motivated by computer aided design and manufacture
  - Boolean operators correspond to mechanical operations.

    E.g., difference is like drilling or milling
  - A common format in CAD products

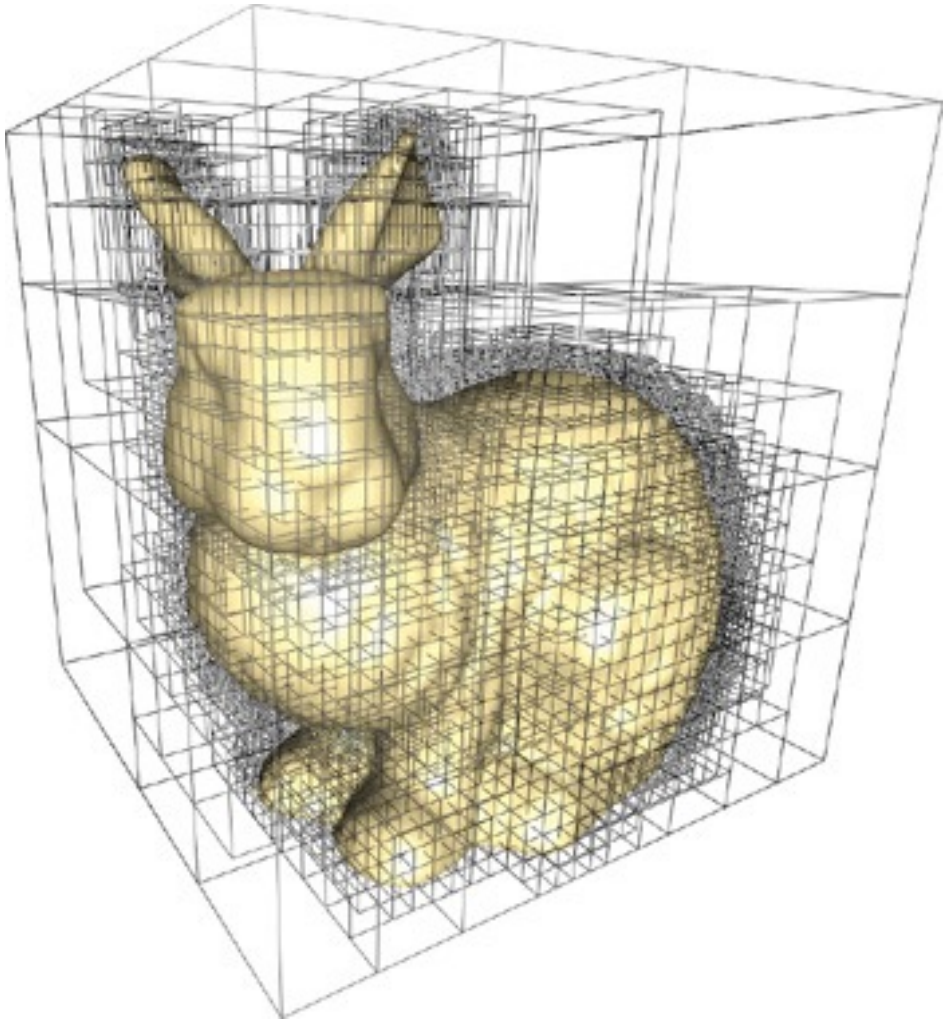
Space subdivision techniques

- Octree representation





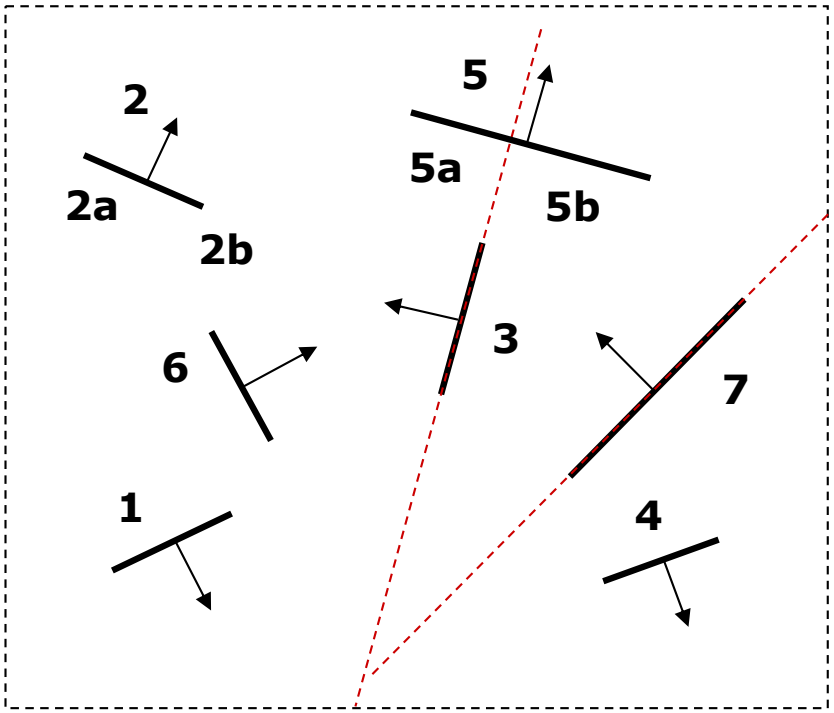
Octree representation

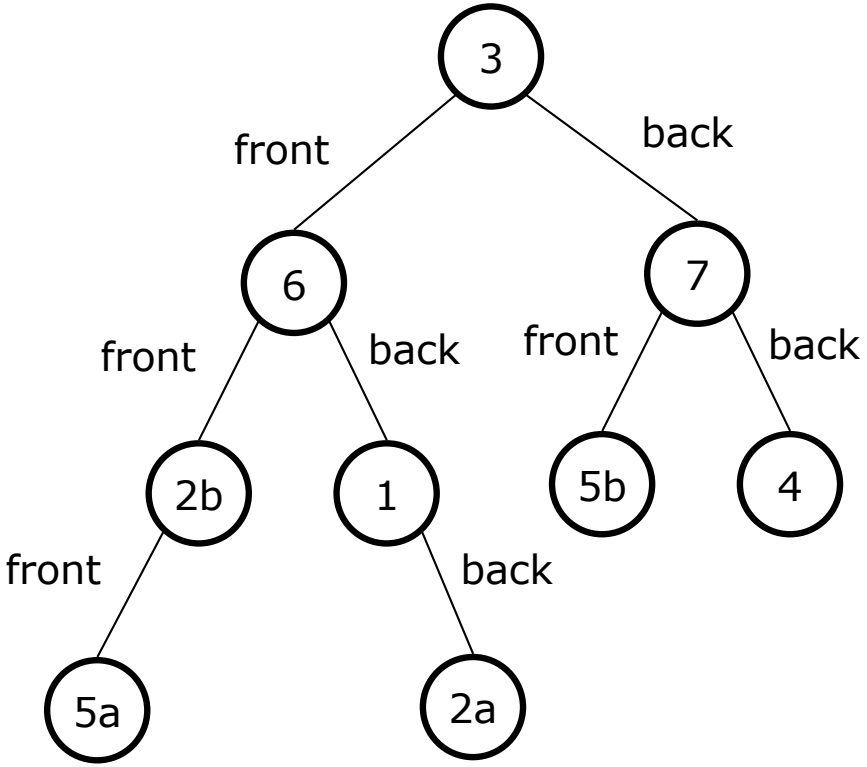


https://developer.nvidia.com/gpugems/GPUGems2/gpugems2_chapter37.html

Space subdivision techniques

- BSP Trees (Binary Space Partitioning Trees)





Procedural modeling

- Fractals, fractal dimension
- Apply algorithmic rules to generate shapes

